FINAL PROJECT

Exploratory Data Analysis Report on Heart Attack Dataset

ALY6010: Probability Theory and Introductory Statistics

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Introduction

For knowing the elements that create heart disease risk helps to establish preventive healthcare measures. This report investigates the primary risks affecting heart attack rate through the statistical methods and hypothesis testing.

The dataset contains patient demographics together with lifestyle factors and physiological measurements. The analysis focuses on testing three hypotheses:

1. Does smoking significantly affect cholesterol levels?

2. The data analysis seeks to determine if heart attack patients show higher or lower blood pressure readings compared to non-heart attack patients.

3. Are men and women at different levels of heart attack risk?

I perform t-tests and chi-square tests to evaluate these hypotheses while using confidence intervals to gain additional understanding.

Hypothesis Testing

Smoking and Cholesterol Levels

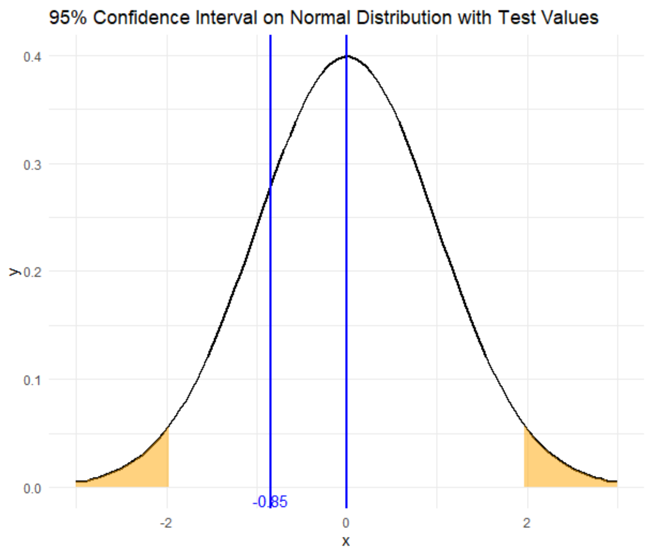
Research Question: Does smoking significantly impact cholesterol levels?

- **H0**: There is no significant difference in mean cholesterol levels between smokers and non-smokers
- **H1**: Smokers have significantly different cholesterol levels than non-smokers

Results:

Name	Type	Value
smoker_cholesterol_test	list [10] (S3: htest)	List of length 10
statistic	double [1]	-0.8463068
parameter	double [1]	372966.2
p.value	double [1]	0.3973822
conf.int	double [2]	-0.530 0.211
estimate	double [2]	199 200
null.value	double [1]	0
stderr	double [1]	0.1890357
alternative	character [1]	'two.sided'
method	character [1]	'Welch Two Sample t-test'
data.name	character [1]	'df$Cholesterol[df$Smoker == 1] and df$Cholesterol[df$Smoker == 0]'

The p-value=0.397 which is greater than 0.05, meanings we fail to reject the null hypothesis. This suggests that smoking does not have a statistically significant impact on cholesterol levels in this dataset.



The curve displays the results from t-tests that explore the relationship between smoking habits and cholesterol levels. The t-statistic value of -0.85 falls within the 95% confidence interval which demonstrates that smoking has no significant impact on cholesterol levels based on this dataset.

Blood Pressure and Heart Attack Outcome

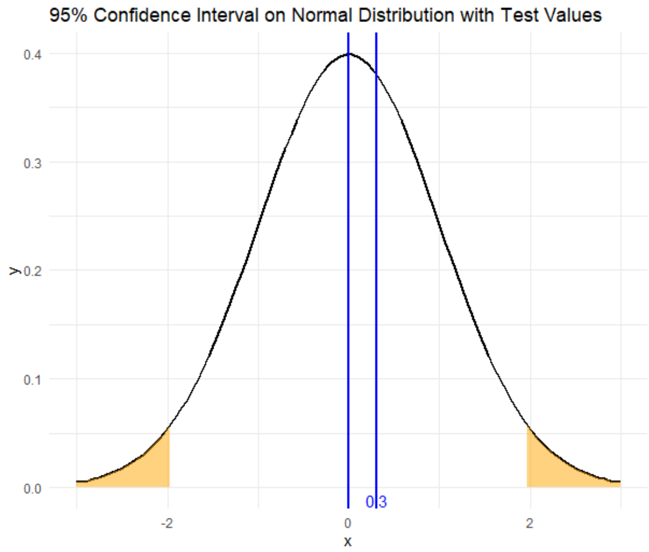
Research Question: Is there a significant difference in blood pressure between heart attack patients and non-heart attack patients?

- **H0:** There is no significant difference in mean blood pressure between heart attack patients and non-heart attack patients.
- **H1:** Heart attack patients have significantly different blood pressure levels than non-heart attack patients.

Results:

Name	Type	Value
bp_test	list [10] (S3: htest)	List of length 10
statistic	double [1]	0.3030909
parameter	double [1]	372964.6
p.value	double [1]	0.7618207
conf.int	double [2]	-0.141 0.193
estimate	double [2]	135 134
null.value	double [1]	0
stderr	double [1]	0.08509564
alternative	character [1]	'two.sided'
method	character [1]	'Welch Two Sample t-test'
data.name	character [1]	'df$BloodPressure[df$Outcome == "Heart Attack"] and df$BloodPressure[df$Outcome ...

Since the p value =0.762 which is greater than 0.05, we fail to reject the null hypothesis. This indicates that blood pressure differences between heart attack patients and non-heart attack patients are not statistically significant.



A t-value of 0.3 suggests minimal difference in mean blood pressure between the two groups because it approaches zero.

	Outcome	Mean_BP	CI_Lower	CI_Upper
1	Heart Attack	134.5211	134.4030	134.6393
2	No Heart Attack	134.4953	134.3776	134.6131

The Confidence Intervals for Blood Pressure by Group show that heart attack patients have a mean blood pressure of 134.52 while non-heart attack patients have a mean blood pressure of 134.50 which demonstrates their values are almost the same. Heart attack patients present 95% confidence intervals that extend from 134.40 to 134.64 while non-heart attack patients present intervals spanning from 134.38 to 134.61 which shows significant overlapping between the two groups. Research findings demonstrate that blood pressure measurements between heart attack patients and non-heart attack patients show no significant statistical distinction.

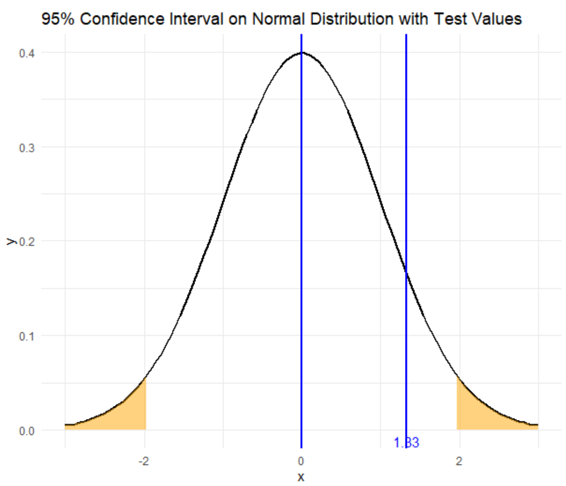
Gender and Heart Attack Risk

Research Question: Do males and females have different heart attack risks?

- **H0:** There is no significant relationship between gender and heart attack occurrences.
- **H1:** Gender has a significant association with heart attack occurrences.

Results:

Name	Type	Value
gender_chisq_test	list [9] (S3: htest)	List of length 9
statistic	double [1]	1.326867
parameter	integer [1]	1
p.value	double [1]	0.2493633
method	character [1]	'Pearson\'s Chi-squared test with Yates\' continuity correction'
data.name	character [1]	'gender_heart_attack_table'
observed	integer [2 x 2] (S3: table)	93193 93123 93011 93647
expected	double [2 x 2]	93017 93299 93187 93471
residuals	double [2 x 2] (S3: table)	0.578 -0.577 -0.578 0.577
stdres	double [2 x 2] (S3: table)	1.16 -1.16 -1.16 1.16



The X^2 = 1.327 is inside the 95% confidence interval, suggesting that there is no strong association between gender and heart attack risk. Since the test statistic does not fall in the extreme ends of the distribution, we fail to reject the null hypothesis, meaning that gender alone is not a significant factor in determining heart attack risk.

Summary

The hypothesis testing shows smoking does not affect cholesterol levels significantly ($p = 0.3974$), blood pressure differences between heart attack patients and non-heart attack patients show no statistical significance ($p = 0.7618$), and gender does not influence heart attack risk in a significant manner ($p = 0.2494$). Blood pressure confidence intervals demonstrate minimal differences between groups which suggests blood pressure cannot effectively distinguish heart attack risks. The findings suggest that while traditional risk factors such as smoking and blood pressure alongside gender play a role in heart disease prediction, they alone do not provide complete insight which implies a need to study additional factors like genetics and lifestyle habits. Heart disease risk evaluation would become more accurate through future research that investigates multiple variable interactions, uses predictive modeling techniques, and applies larger and more diverse datasets.

References

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